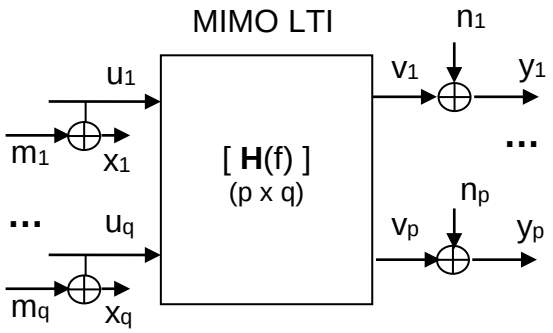
Non-parametric SYSTEM identification of MIMO sytems - Intro

MIMO-system with input & output measurement noise



- MIMO system outputs $v_1, v_2, \ldots, v_p$ are all generated by summing of linearly filtered MIMO system inputs $u_1, u_2, \ldots, u_q$
- The MIMO system is characterized by an $(p \times q)$ transfer function matrix $\mathbf{H}(f)$
- $m_1, m_2, \ldots, m_q$ are the input measurement noises
- $n_1, n_2, \ldots, n_p$ are the output measurement noises
- $x_1, x_2, \ldots, x_q$ are the noisy input measurements
- $y_1, y_2, \ldots, y_p$ are the noisy output measurements

Case 1: only <u>output</u> measurement noise, $\forall\, \mathbf{m_i} = \mathbf{0}$

$$\underset{(p\times 1)}{\mathbf{Y}(f)} = \underset{(p\times q)}{\mathbf{H}(f)}\underset{(q\times 1)}{\mathbf{X}(f)} + \underset{(p\times 1)}{\mathbf{N}(f)}$$

Postmultiplying by $\mathbf{X}^{H}(f)$ and taking expectation gives

$$\underset{(p\times q)}{\mathbf{G}_{xy}(f)} = \underset{(p\times q)}{\mathbf{H}(f)}\underset{(q\times q)}{\mathbf{G}_{xx}(f)} + \underset{(p\times q)}{\mathbf{0}} \qquad \text{and}$$

$$\underset{(p\times q)}{\widehat{\mathbf{H}}_1(f)} = \underset{(p\times q)}{\mathbf{G}_{xy}(f)}\underset{(q\times q)}{\mathbf{G}_{xx}^{-1}(f)}$$

Case 2: only <u>input</u> measurement noise, $\forall\ \mathbf{n_i} = \mathbf{0}$

$$\underset{(p\times 1)}{\mathbf{Y}(f)} = \underset{(p\times q)}{\mathbf{H}(f)}\underset{(q\times 1)}{[\mathbf{X}(f)} - \underset{(q\times 1)}{\mathbf{M}(f)}]$$

Postmultiplying by $\mathbf{Y}^{H}(f)$ and taking expectation gives

$$\underset{(p\times p)}{\mathbf{G}_{yy}(f)} = \underset{(p\times q)}{\mathbf{H}(f)}\underset{(q\times p)}{\mathbf{G}_{yx}(f)} - \underset{(p\times p)}{\mathbf{0}}$$

Note that the necessary matrix inversion is only possible if the number of inputs equals the number of outputs, $p = q$ giving

$$\underset{(p\times p)}{\widehat{\mathbf{H}}_2(f)} = \underset{(p\times p)}{\mathbf{G}_{yy}(f)}\underset{(p\times p)}{\mathbf{G}_{yx}^{-1}(f)}$$